

QXU5010

Surface Silanisation and Contact Angle Goniometry – Coursework Instructions

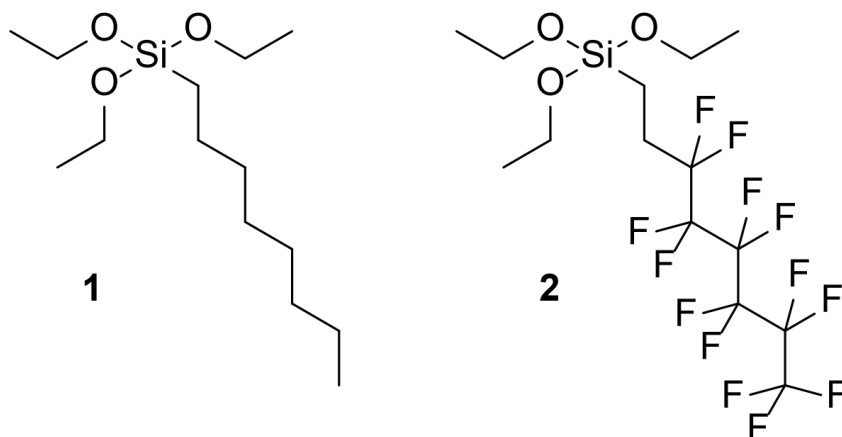
November 2025 – Semester 1 – Week 11

Report due on Thursday 12th December at 11:59 pm. Maximum 5 sides of A4 in total, font 10 or above, single line spacing. Include your group name and the title of this coursework in the header or footer.

1. Objectives of the Project

Surface chemical modification can completely change the surface properties of materials. In particular, monolayers formed on surfaces can be used to control the hydrophilic/hydrophobic character of materials and their wettability. This has important applications in the formulation of composites, the processing of materials in thin layers, but also the patterning of proteins or cells for biotechnologies. Glass and silicon oxide substrates can conveniently be functionalised with silane monolayers, *via* a process known as silanisation. This involves the reaction of silicon oxide surfaces with chlorosilane or alkoxy silane molecules forming monolayers at their surface. Such monolayers can be designed to display a wide range of hydrophilic/hydrophobic properties, with water contact angles ranging from 0° to 120°.

In this coursework task, you will author a report based on a series of surfaces prepared *via* silanisation and characterise their surface chemistry by the examination of contact angle goniometry data generated in the lab. These experiments were performed using small silicon substrates (1x1 cm approximately), which are plasma treated before reacting them with triethoxysilanes 1 and 2. The surfaces were allowed to react for at least 15 minutes. You will measure the contact angles of three different liquids (water, ethylene glycol, and toluene) at these surfaces and analyse the data.



2. Experimental Protocol [Provided in Full Detail in the Lab Worksheet Document]

These steps were followed in the preparation of the surfaces and characterisation of their surface chemistry:

1. **Plasma treatment of silicon substrates.** Silicon chips were placed face up (the side that appears reflective) in the plasma oven. The chips were plasma treated for 5 mins with 100% intensity (100 W Power, air pressure of 1.1×10^{-2} Torr). During the treatment, the plasma chamber illuminates with a characteristic pink colour for air (O₂) plasma.
2. **Formation of silane monolayers.** After plasma treatment, the samples were directly placed in the silane solutions (a separate solution was used for each silane, i.e.; 1 and 2). A sample

was retained from the plasma treatment that would be directly characterised by contact angle goniometry for comparison.

3. **Surface characterisation.** Images of contact angles of three different liquids (water, ethylene glycol, toluene) at the surface of the three samples prepared (plasma-treated, functionalised with silane 1 and functionalised with silane 2) were taken. Each group of students will analyse these images, once contact angle measurements are carried out.

[Note: two sets of images will be provided, both sets can be analysed by each group].

3. Preparation of the Report

Your group's report must include the following:

- A **one-page introduction** of surface tension and wetting will then be provided, this must include references to the scientific literature.
- This is followed by an indication/description of your **analysis method**. Specifically, include an image of a droplet and show how the angles were assigned.
- In addition, plot $\cos(\theta) = f(\gamma)$ for the set of data you obtained (plot data obtained for all three surfaces on the same graph), where θ is the angle made between the surface and the contact line of the droplet and γ is the surface tension of the liquids you have been using (water, ethylene glycol and dodecane – obtained from the literature). This is the **Zisman plot** (see *workshop lecture notes*). Then **determine the critical surface tension** of each surface.
- Include a **brief reflection** (up to one page) describing how the lab session went, noting any unexpected observations or challenges. Summarise the positive and negative aspects of the experiment and its workflow. Conclude with suggestions for future experiments or variations you think would be valuable to explore.

To summarise, please include the following sections; **One-page Introduction of Surface Tension, Image Analysis, Zisman Plot (Critical Surface Tensions), and Lab Reflections.**

3.1 Finally include answers to the following questions:

1. Draw the chemical structures of the three surfaces you studied (plasma-treated silicon, monolayers of silanes 1 and 2 on silicon).
2. List the surfaces in order of their critical surface tension, rationalise the relative values.
3. Describe the surface localised reaction that takes place during the functionalisation using silanes 1/2. Provided this, theorise an experimental change that would increase the amount/density of silane attached to the surface.

4. Peer review

A review of your individual performance will be made by your peers (group) and used to calculate the final grade. If all group members contribute equally, all will receive the final mark given for the report. Groups also have the option to include a summary/breakdown of group member report contributions. This could be used to support peer review scores and in cases of academic misconduct.

5. Additional notes:

- A mark scheme is provided, this will indicate how your group's report will be assessed.
- The format of the report (e.g. order of sections) is flexible, other than that outlined above.
- All the sections must fit into five pages. Groups can include a cover page, this, as well as reference/bibliography and contribution summary, which are not included in the page count.
- Please use the QMPlus forum for any questions relating to this coursework.